

Benchmark Method Specifications

Framework	Core Parameters	Specification / Baseline Setup
HERA	Threshold Bootstrap (B = 2000)	Sequential Hierarchical (M1 -> M2 -> M3); Holm-Bonferroni FWER control (alpha = 0.05)
TOPSIS	Criteria Weights: w = [0.40, 0.40, 0.20]	Vector Normalization; Euclidean distance to ideal solutions; relative closeness ranking
Borda Count	Positional Rank Sum (Equal weights across M1-M3)	Sum of individual metric ranks; consensus ranking by lowest rank sum
Wilcoxon-Copeland	Pairwise Tournament (Net wins minus losses)	Paired Wilcoxon test with step-down Holm-Bonferroni FWER control (alpha = 0.05)

Experimental Benchmark Scenarios (24 Conditions)

Benchmark Suite	N Conditions	Candidates (N)	Sample Size (n)	Noise (σ , %)	Effect Size (Cliff's d [Range])	Correlation (ρ)
Core	15	10, 12, 14	50	2%, 4%, 6%, 8%, 10%	$d \approx 0.50$ [0.45 - 0.60]	0
Sample Size Sensitivity	6	10, 12, 14	25, 100	4%	$d \approx 0.50$ [0.45 - 0.60]	0
Effect Magnitude Sensitivity	2	12	50	4%	$d \approx 0.25$ [0.20 - 0.30], $d \approx 0.80$ [0.75 - 0.90]	0
Correlation Sensitivity	1	12	50	4%	$d \approx 0.50$ [0.45 - 0.60]	0.5

Ground-Truth Benchmark Calibration

Candidate Hierarchy (N = 10)

Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	74.0%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	70.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

Candidate Hierarchy (N = 12)

Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	74.7%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	73.3%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	70.7%	75.0%	80.0%	Monotonic Baseline Step
C11 (Rank 11)	69.3%	75.0%	80.0%	Monotonic Baseline Step
C12 (Rank 12)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

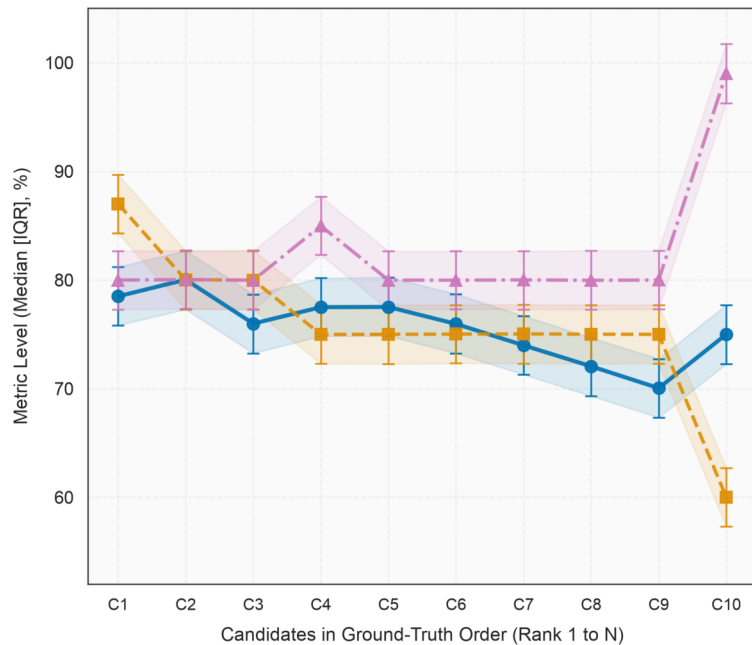
Candidate Hierarchy (N = 14)

Candidate	Metric M1	Metric M2	Metric M3	Benchmarking Role
C1 (Rank 1)	78.5%	87.0%	80.0%	Multi-Criteria Leader (M2 Margin)
C2 (Rank 2)	80.0%	80.0%	80.0%	Primary Metric Leader (Highest M1)
C3 (Rank 3)	76.0%	80.0%	80.0%	Secondary Metric Margin (M2 Step)
C4 (Rank 4)	77.5%	75.0%	85.0%	Tertiary Metric Margin (M3 Step)
C5 (Rank 5)	77.5%	75.0%	80.0%	Multi-Criteria Baseline (M1–M3)
C6 (Rank 6)	76.0%	75.0%	80.0%	Monotonic Baseline Step
C7 (Rank 7)	75.0%	75.0%	80.0%	Monotonic Baseline Step
C8 (Rank 8)	74.0%	75.0%	80.0%	Monotonic Baseline Step
C9 (Rank 9)	73.0%	75.0%	80.0%	Monotonic Baseline Step
C10 (Rank 10)	72.0%	75.0%	80.0%	Monotonic Baseline Step
C11 (Rank 11)	71.0%	75.0%	80.0%	Monotonic Baseline Step
C12 (Rank 12)	70.0%	75.0%	80.0%	Monotonic Baseline Step
C13 (Rank 13)	69.0%	75.0%	80.0%	Monotonic Baseline Step
C14 (Rank 14)	75.0%	60.0%	99.0%	Asymmetric Profile (Low M2, High M3)

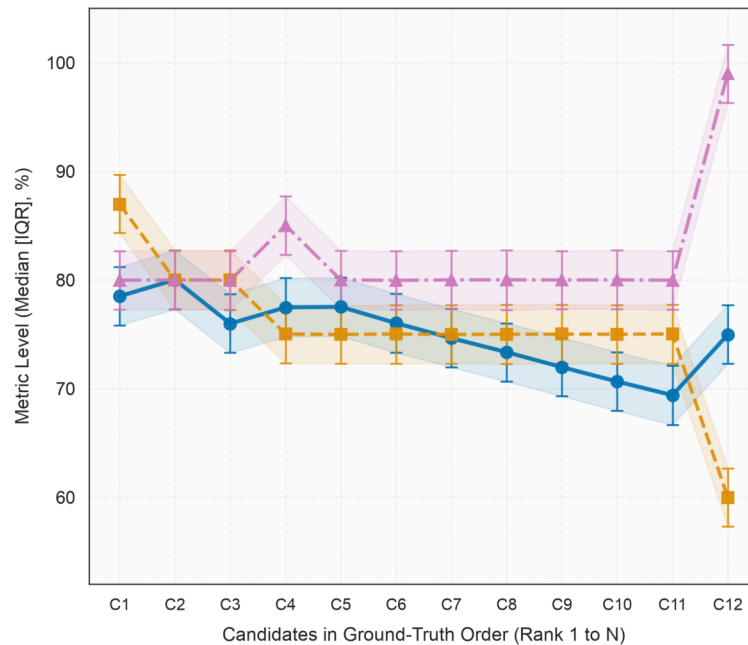
Ground-Truth Candidate Profiles

Candidate input presentation order was randomly permuted across trials to eliminate positional bias

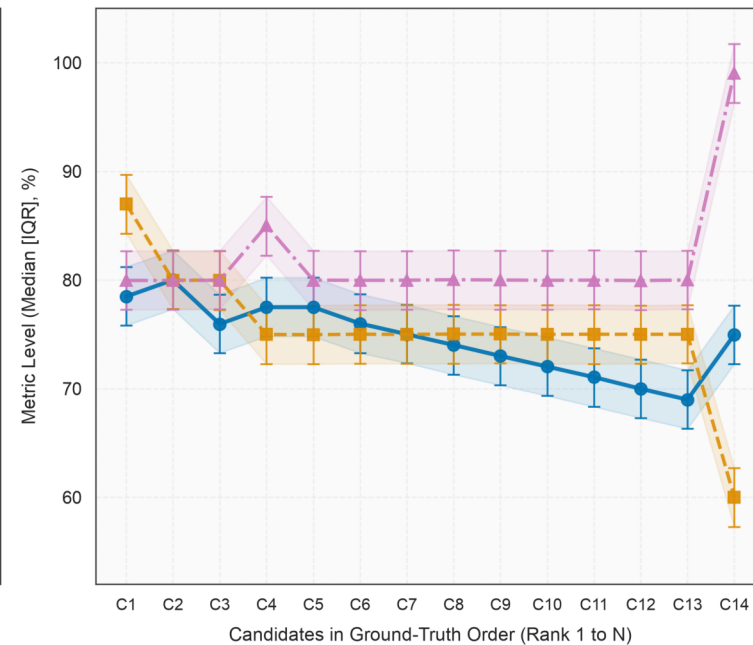
Candidate Profiles (N = 10)



Candidate Profiles (N = 12)



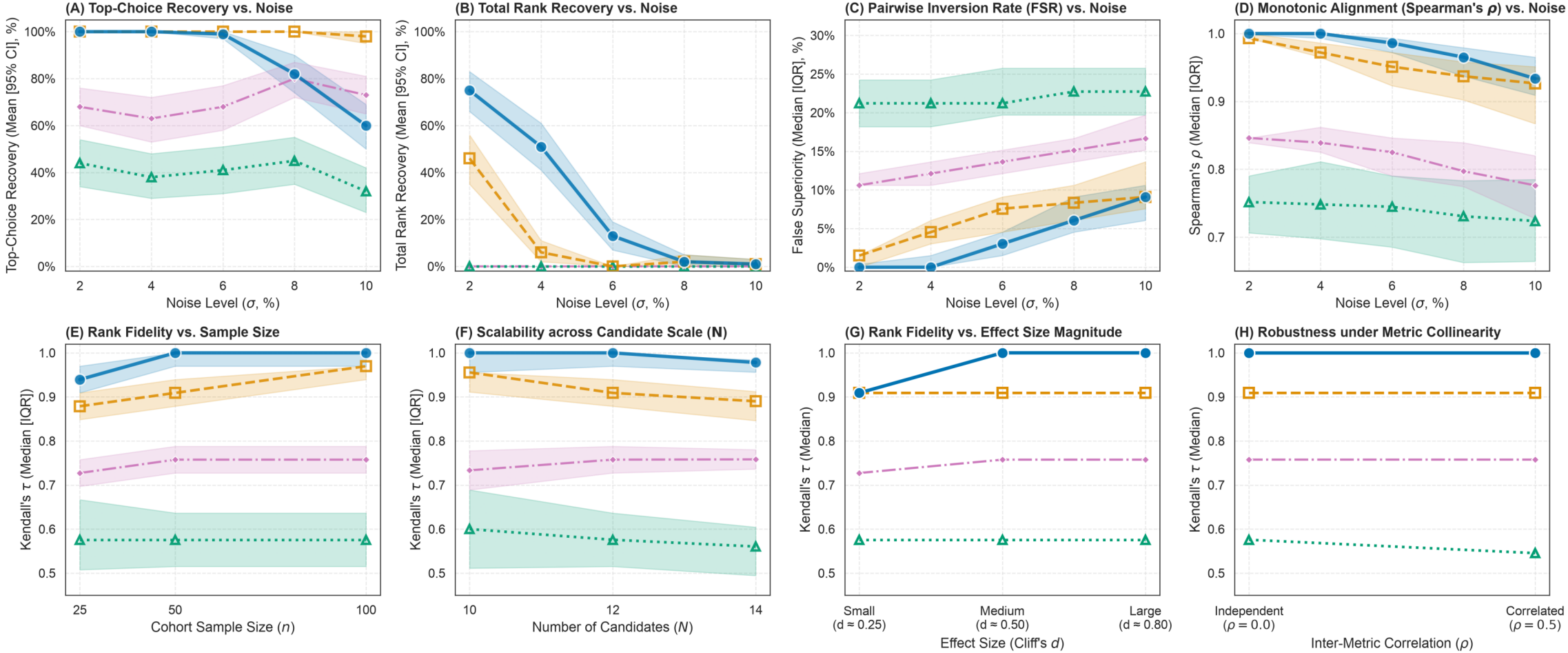
Candidate Profiles (N = 14)



M1 (Primary Metric) M2 (Secondary Metric) M3 (Tertiary Metric)

Benchmark Validation Overview

Default parameters (unless otherwise specified): $N = 12$, $n = 50$, Cliff's $d \approx 0.50$, $\rho = 0.0$, $K = 100$ runs per point

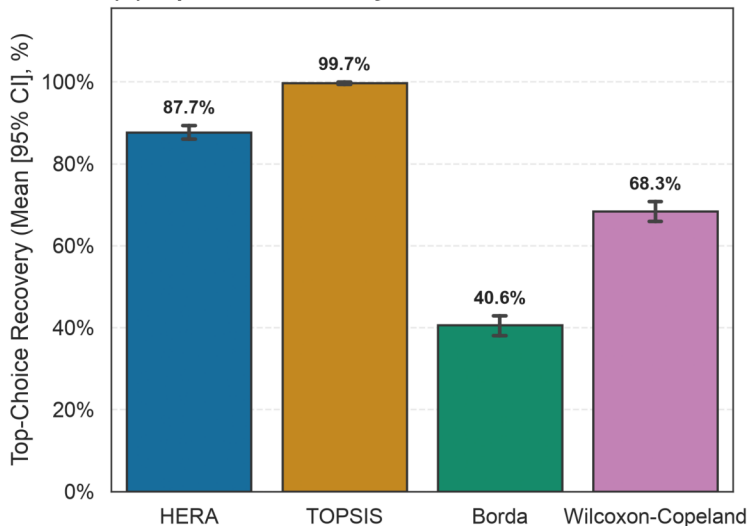


HERA TOPSIS Borda Wilcoxon-Copeland

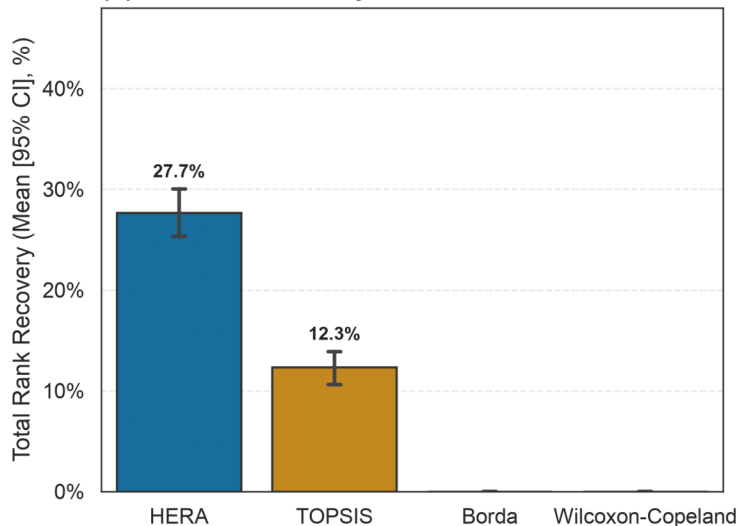
Pooled Distribution across Core Conditions

$N \in \{10, 12, 14\}$, $\sigma \in \{2\%, 4\%, 6\%, 8\%, 10\%\}$, $n = 50$, $\rho = 0.0$; $K = 1500$ runs per method

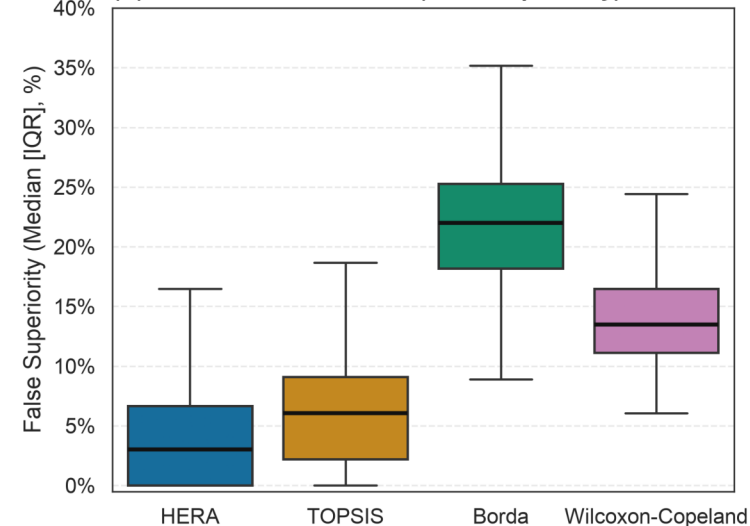
(A) Top-Choice Recovery Rate



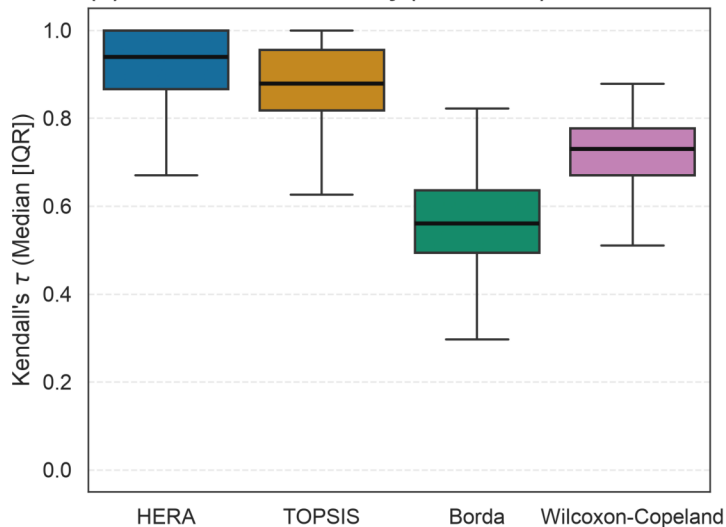
(B) Total Rank Recovery



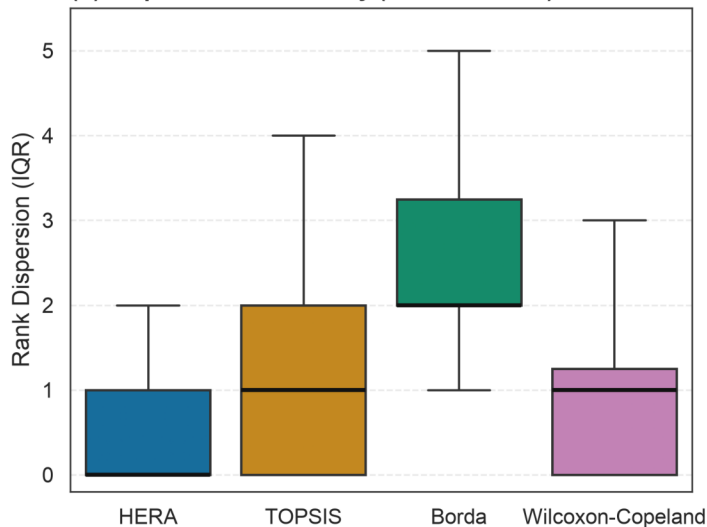
(C) Pairwise Inversion Rate (False Superiority)



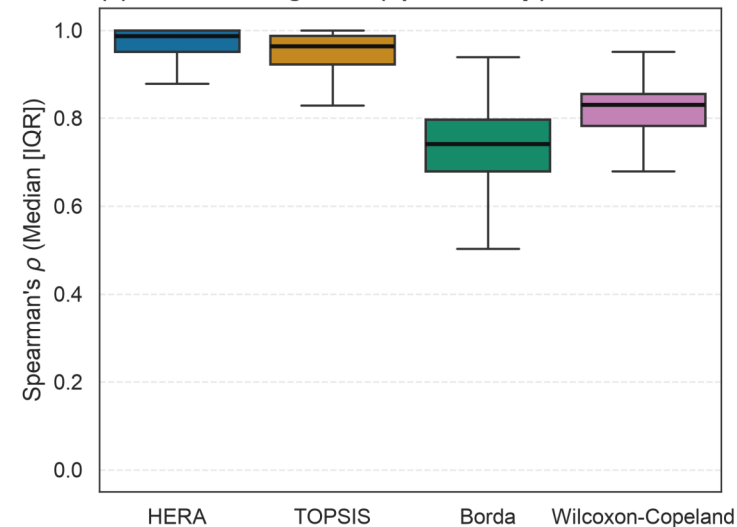
(D) Rank Correlation Fidelity (Kendall's τ)



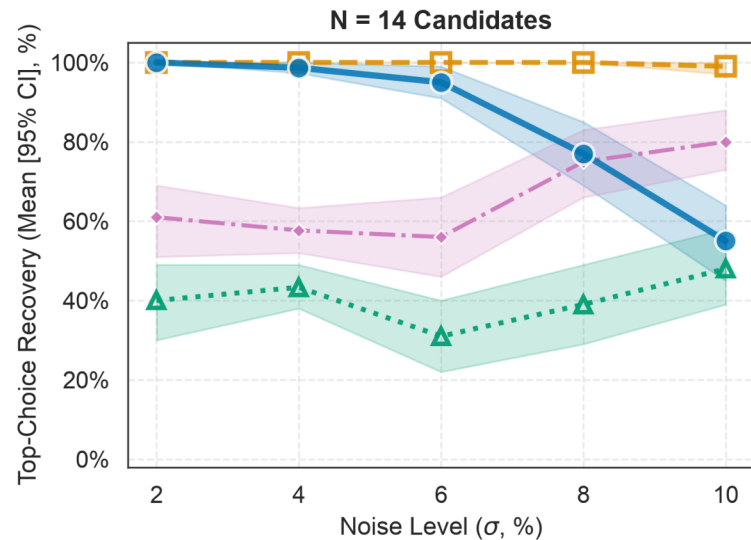
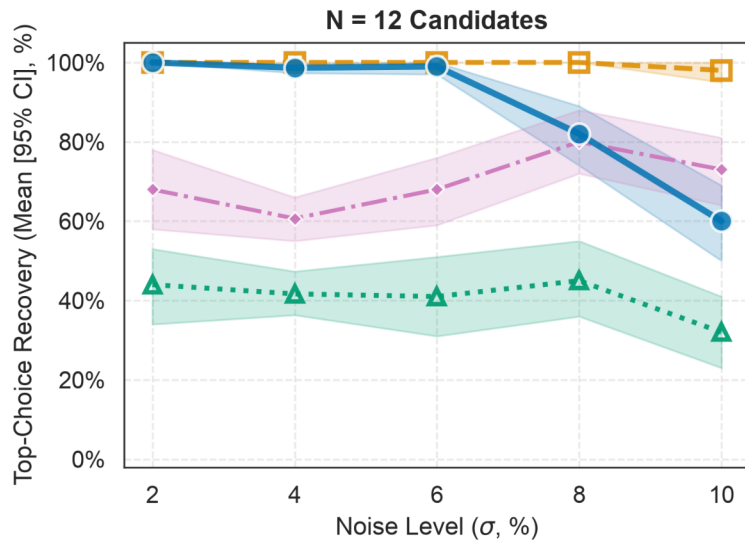
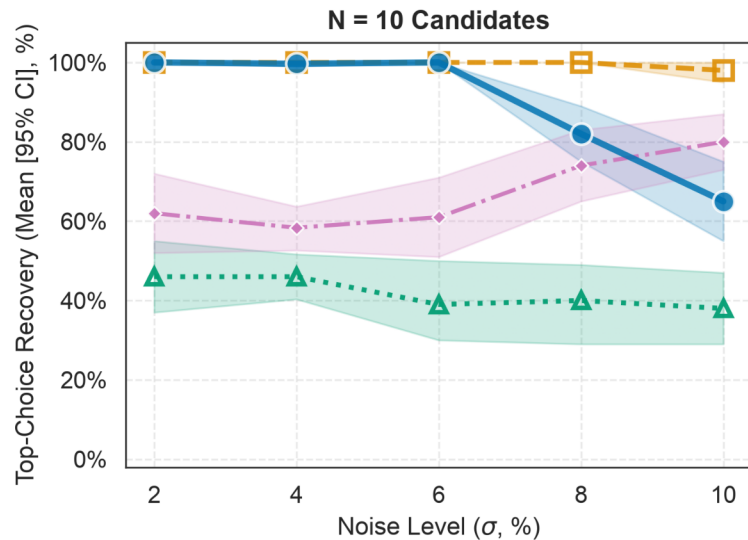
(E) Empirical Rank Stability (Candidate-IQR)



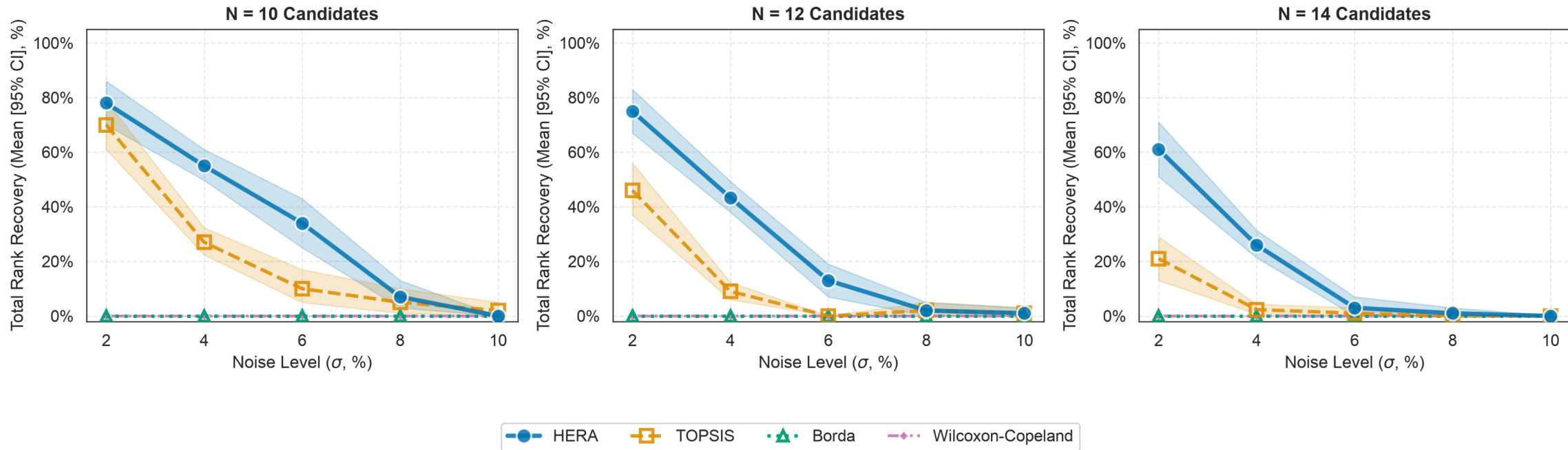
(F) Monotonic Alignment (Spearman's ρ)



Top-Choice Recovery Rate vs. Noise Level
 $N \in \{10, 12, 14\}$, $n = 50$, $\rho = 0.0$, $K = 100$ runs per point

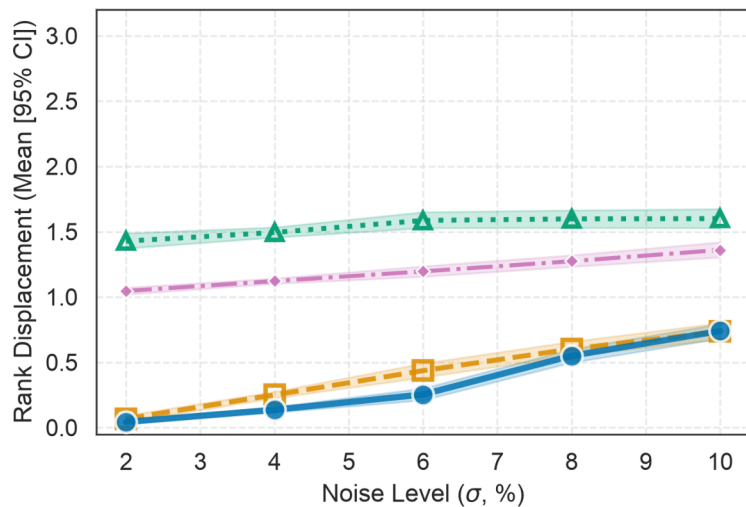


Total Rank Recovery vs. Noise Level
 $N \in \{10, 12, 14\}$, $n = 50$, $\rho = 0.0$, $K = 100$ runs per point

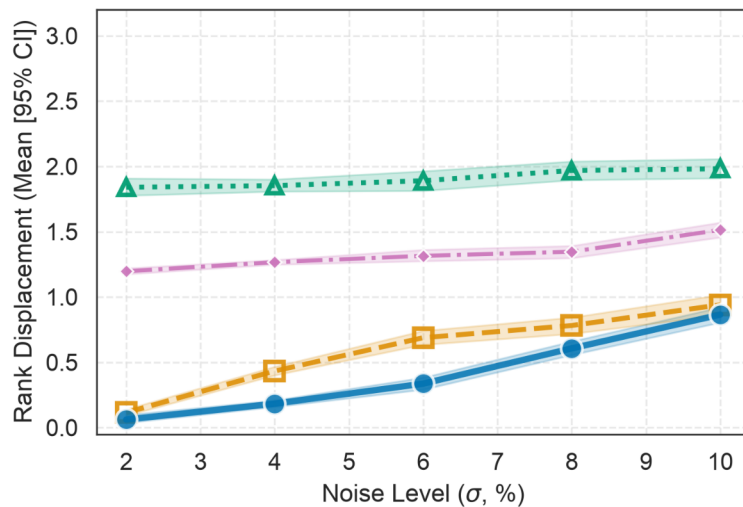


Mean Rank Displacement vs. Noise Level
 $N \in \{10, 12, 14\}$, $n = 50$, $\rho = 0.0$, $K = 100$ runs per point

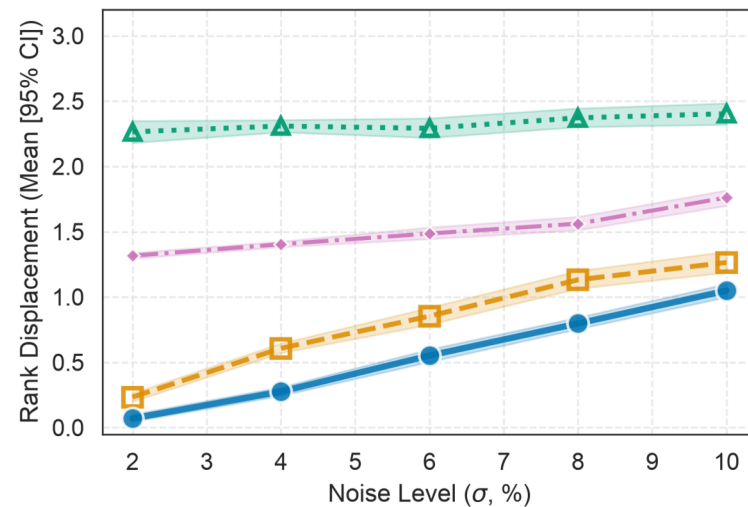
N = 10 Candidates



N = 12 Candidates

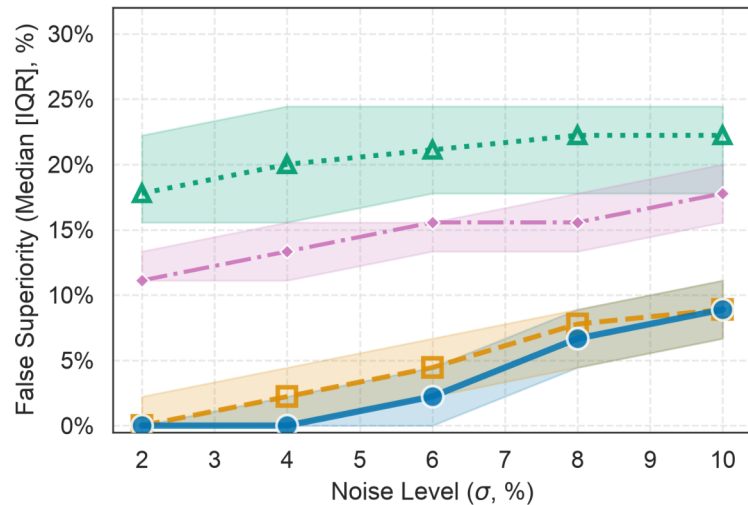


N = 14 Candidates

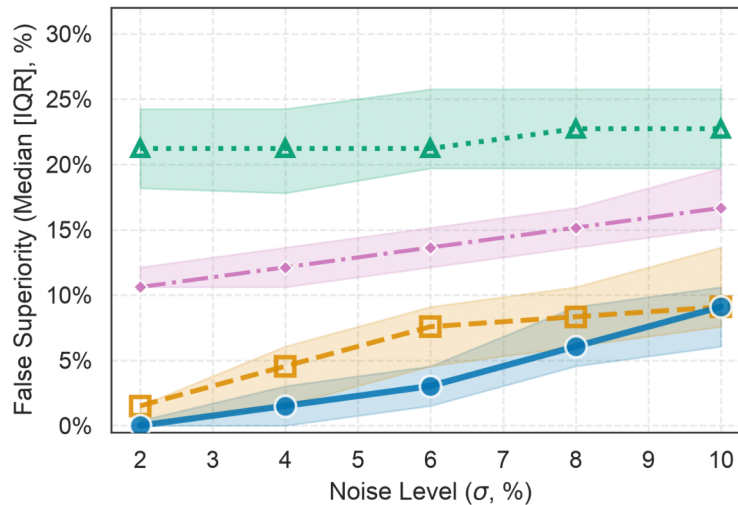


Pairwise Inversion Rate (FSR) vs. Noise Level
 $N \in \{10, 12, 14\}$, $n = 50$, $\rho = 0.0$, $K = 100$ runs per point

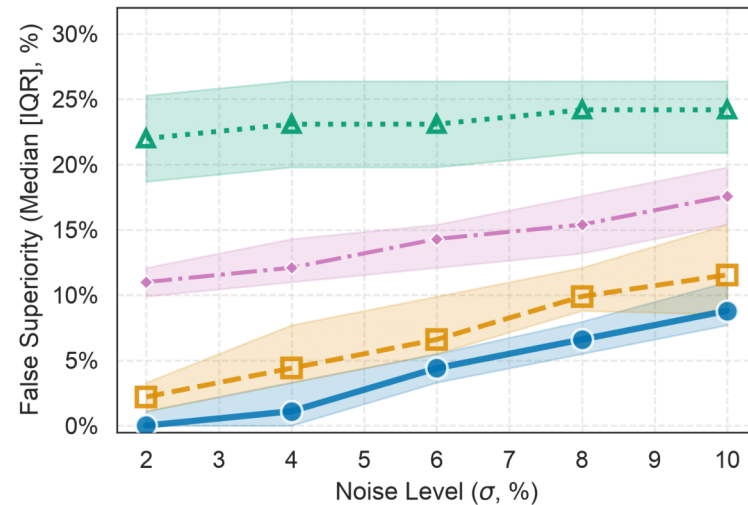
N = 10 Candidates



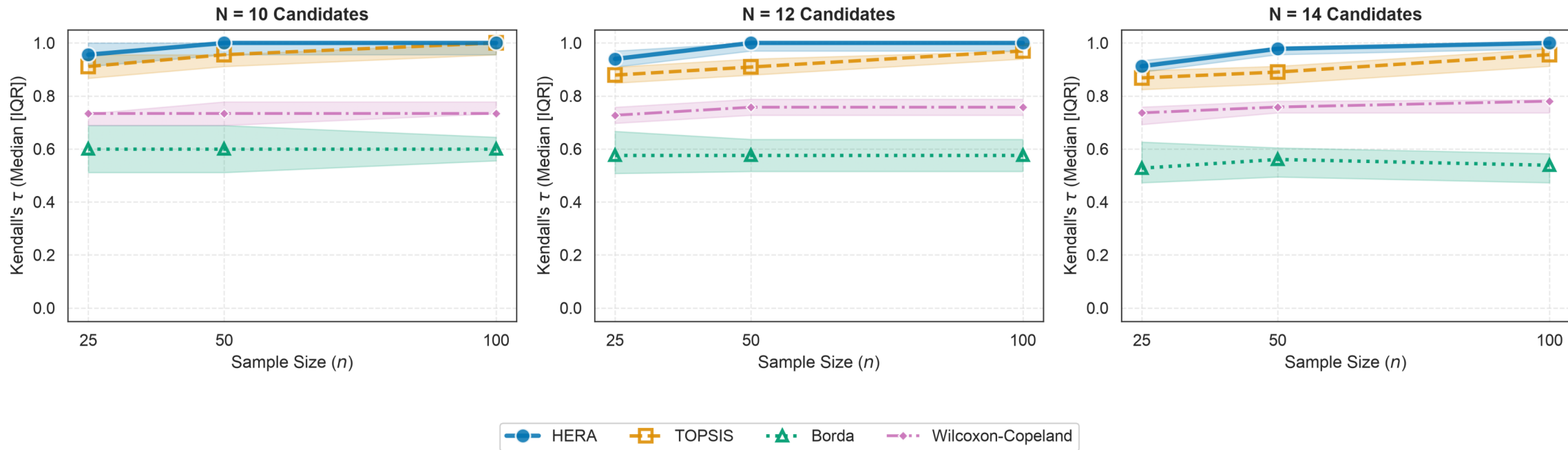
N = 12 Candidates



N = 14 Candidates

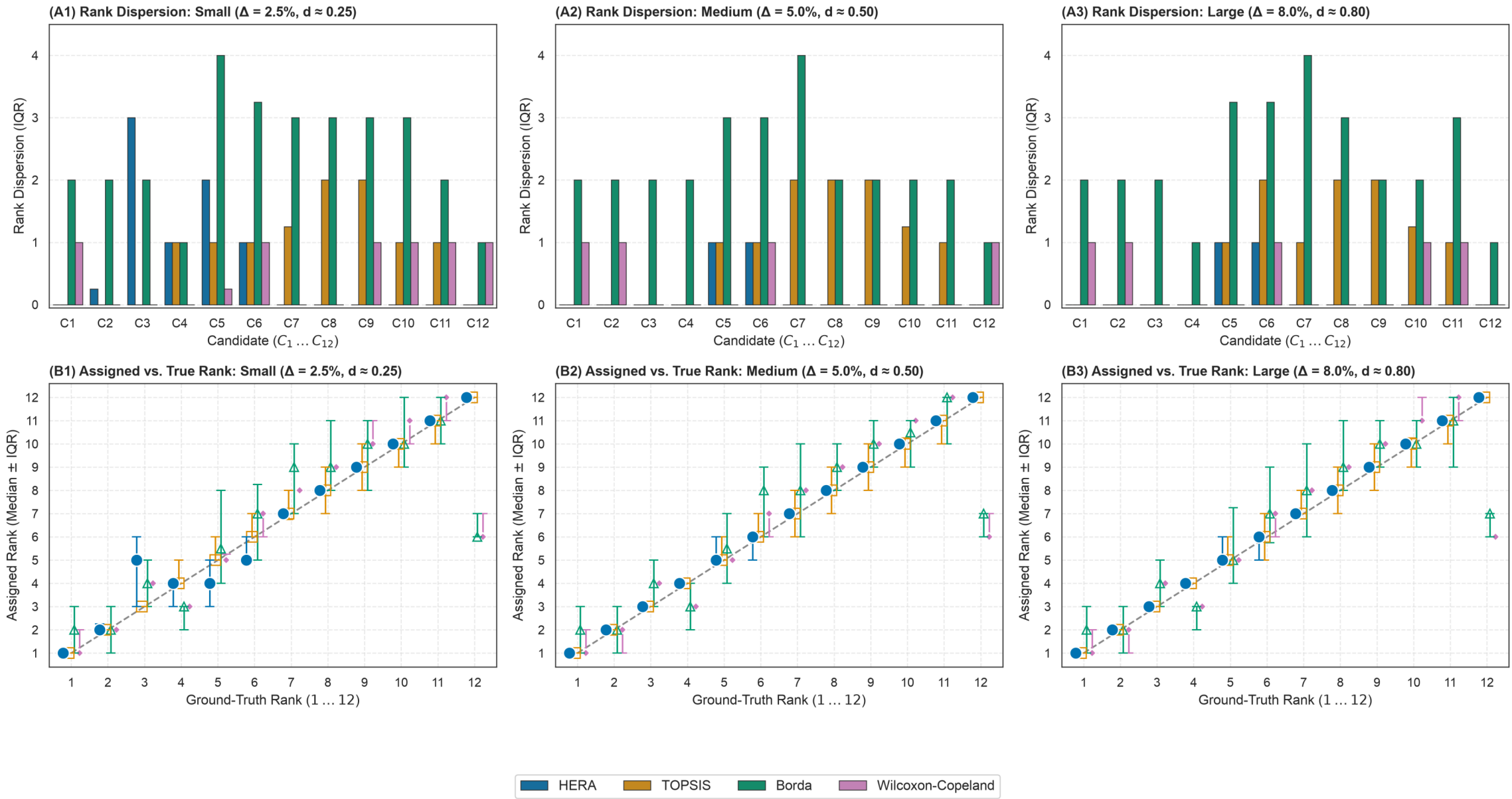


Rank Correlation (Kendall's τ) vs. Sample Size
 $N \in \{10, 12, 14\}$, $\sigma = 4\%$, $\rho = 0.0$, $K = 100$ runs per point



Effect Size Sensitivity: Candidate-Level Rank Dispersion & Fidelity

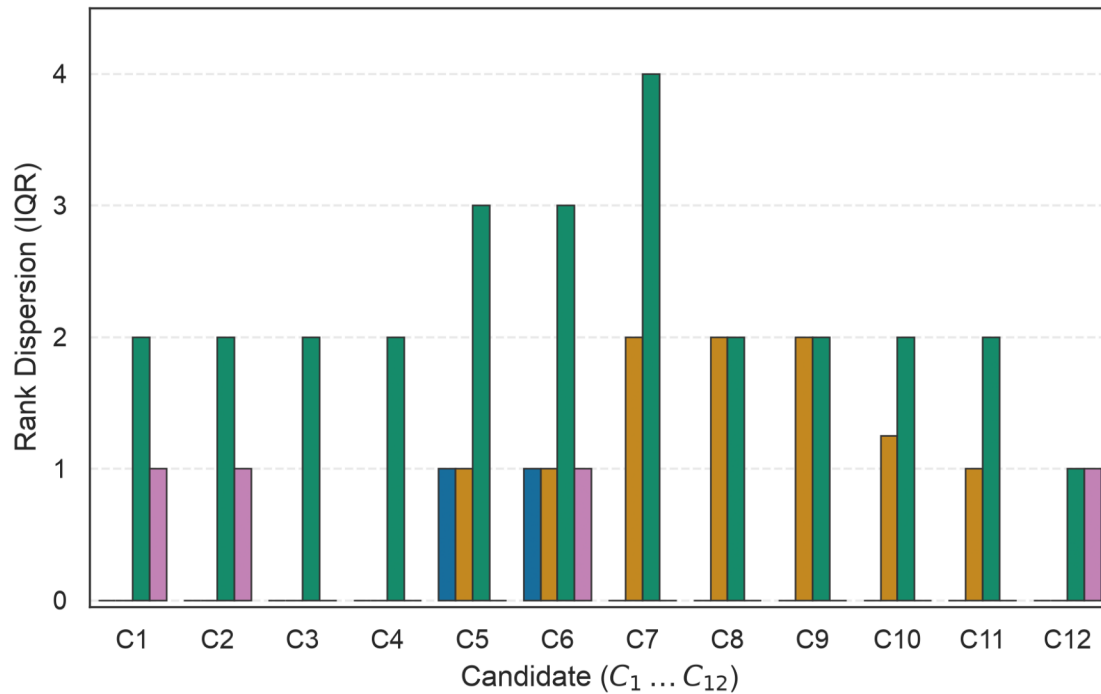
N = 12, n = 50, $\sigma = 4\%$, $\rho = 0.0$, K = 100 runs per condition



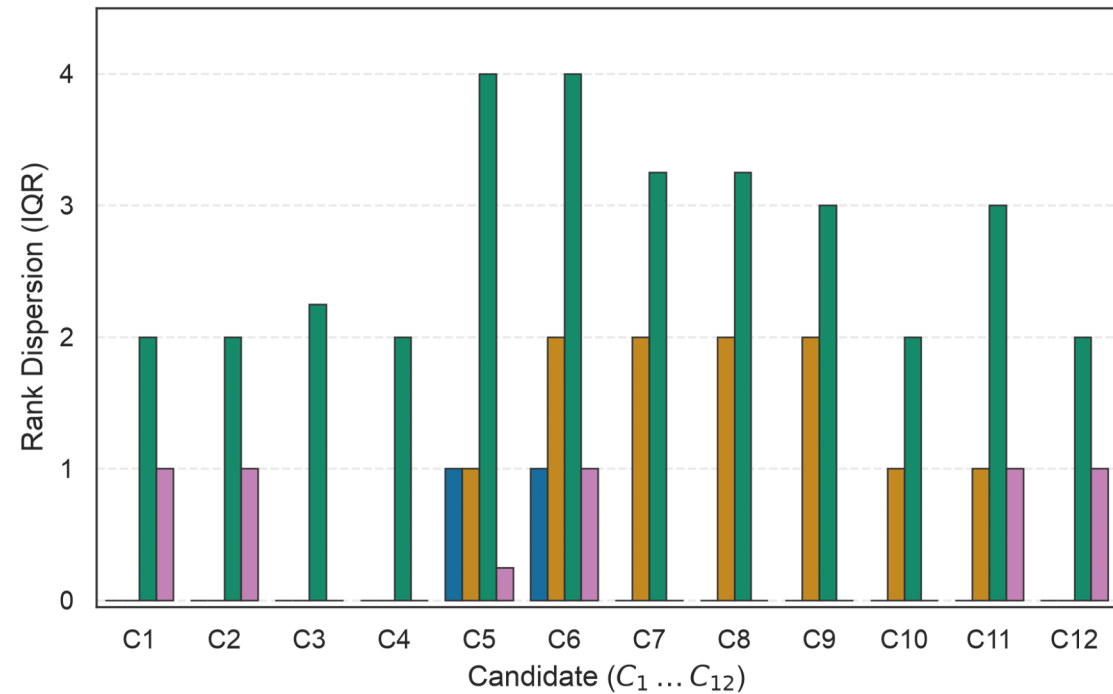
Inter-Metric Collinearity Sensitivity: Rank Dispersion & Fidelity

$N = 12$, $n = 50$, $\sigma = 4\%$, Cliff's $d \approx 0.50$, $K = 100$ runs per condition

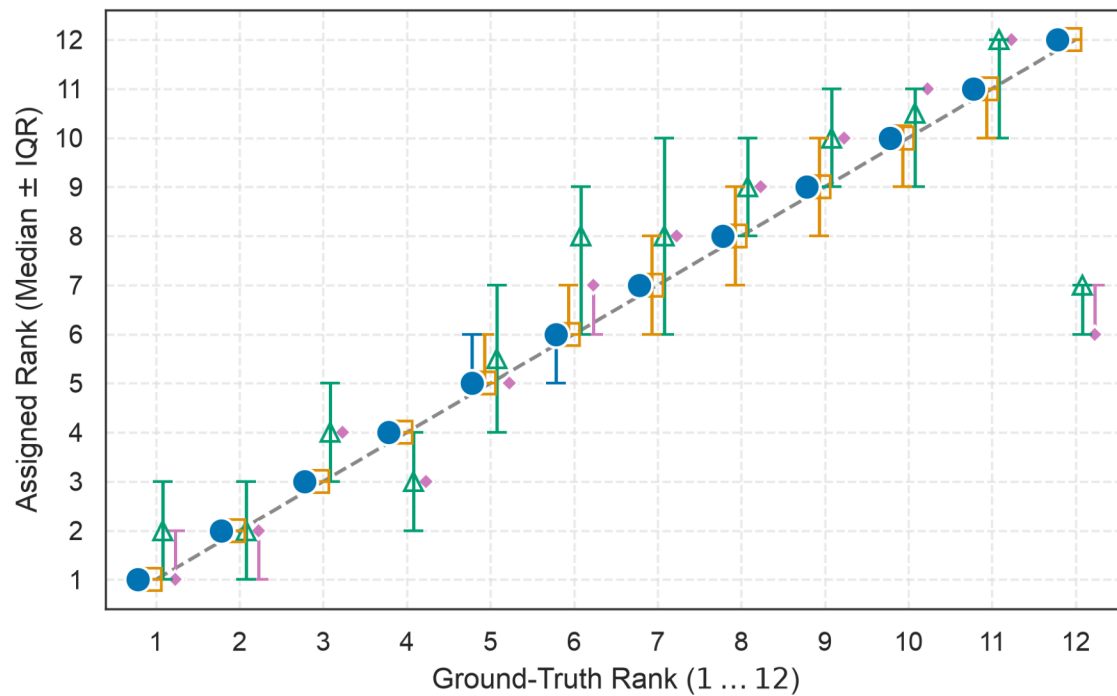
(A1) Rank Dispersion: Independent Criteria ($\rho = 0.0$)



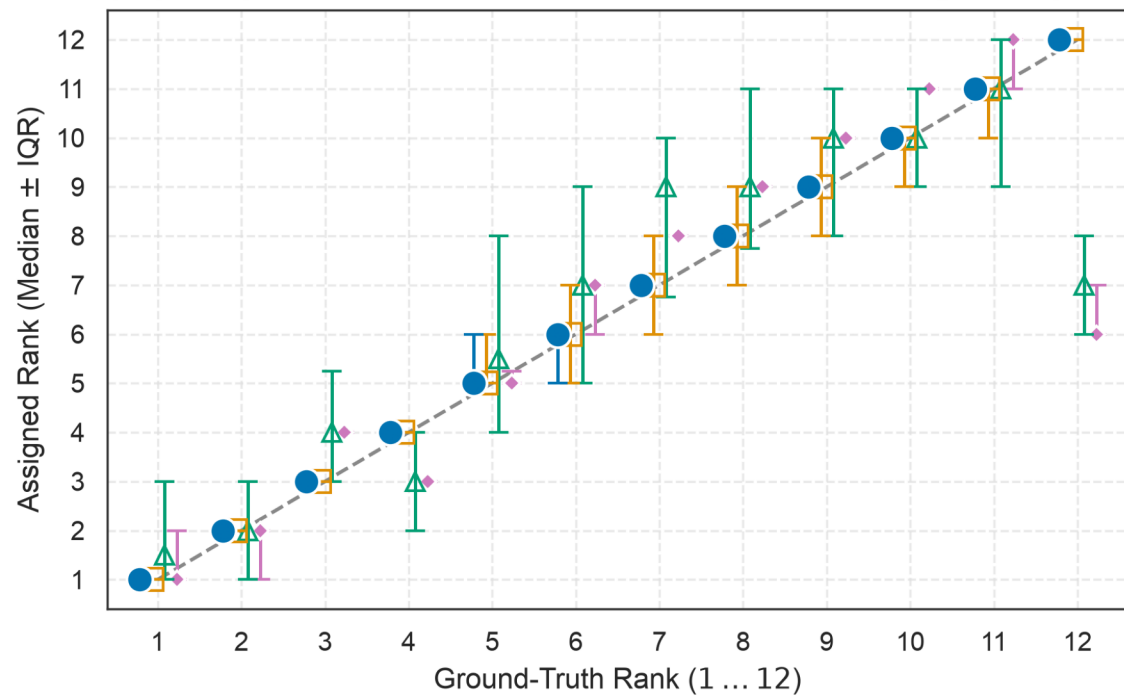
(A2) Rank Dispersion: Correlated Criteria ($\rho = 0.5$)



(B1) Assigned Rank vs. Ground Truth: Independent Criteria ($\rho = 0.0$)



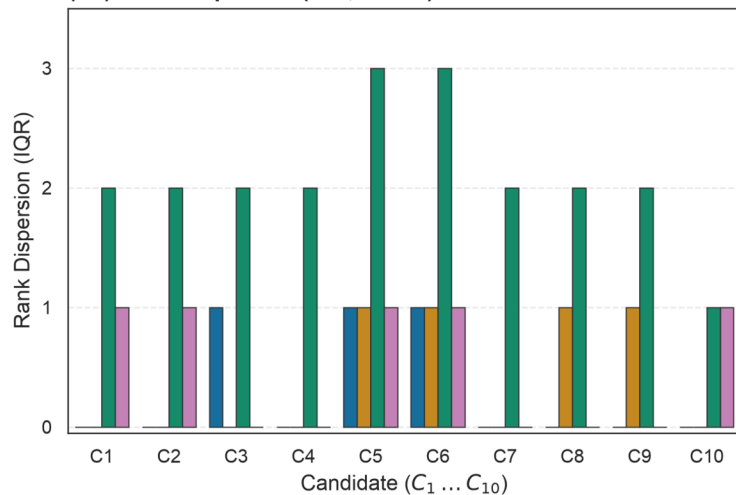
(B2) Assigned Rank vs. Ground Truth: Correlated Criteria ($\rho = 0.5$)



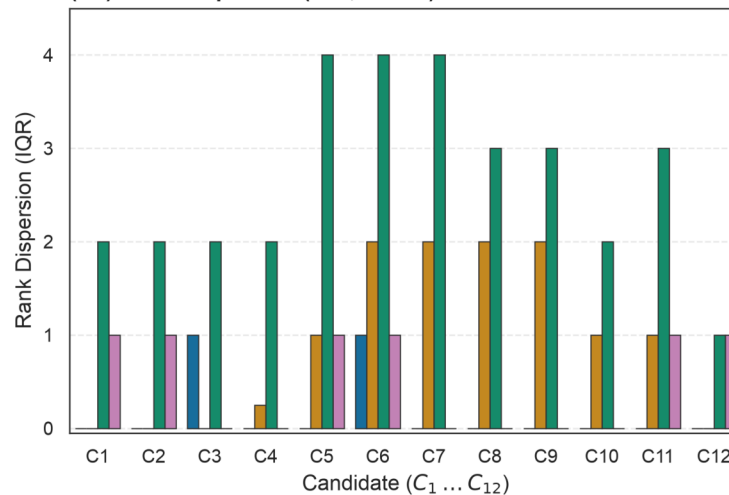
HERA TOPSIS Borda Wilcoxon-Copeland

Candidate-Level Rank Stability Across Scales
 $N \in \{10, 12, 14\}$, $n = 50$, $\sigma = 4\%$, $\rho = 0.0$, $K = 100$ runs per point

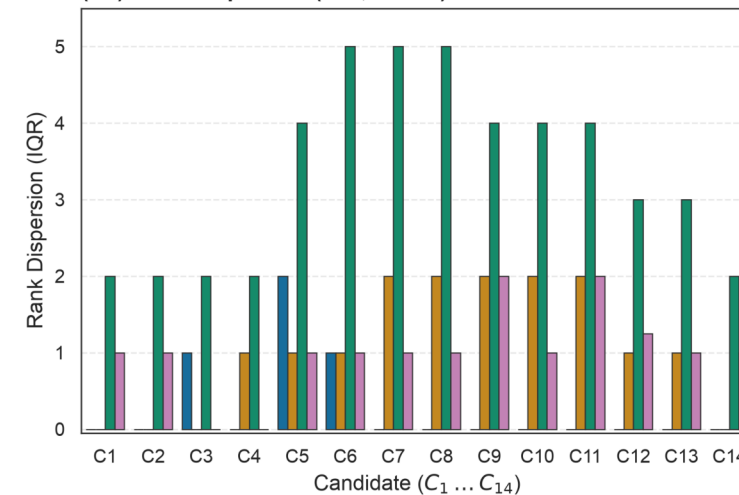
(A1) Rank Dispersion (IQR, $N = 10$)



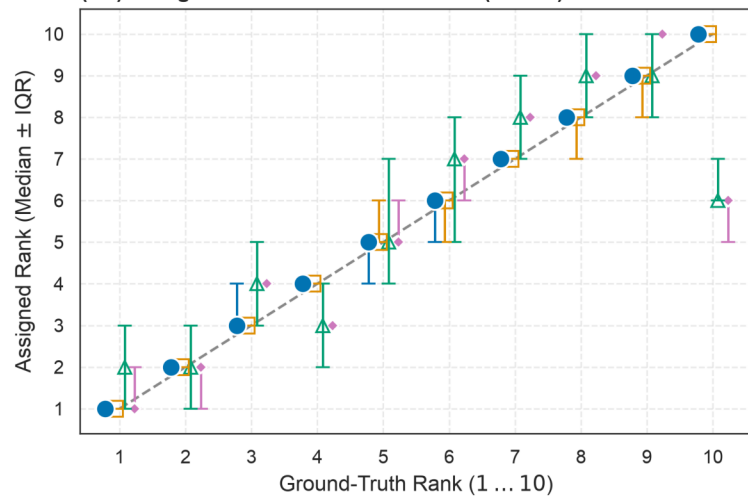
(A2) Rank Dispersion (IQR, $N = 12$)



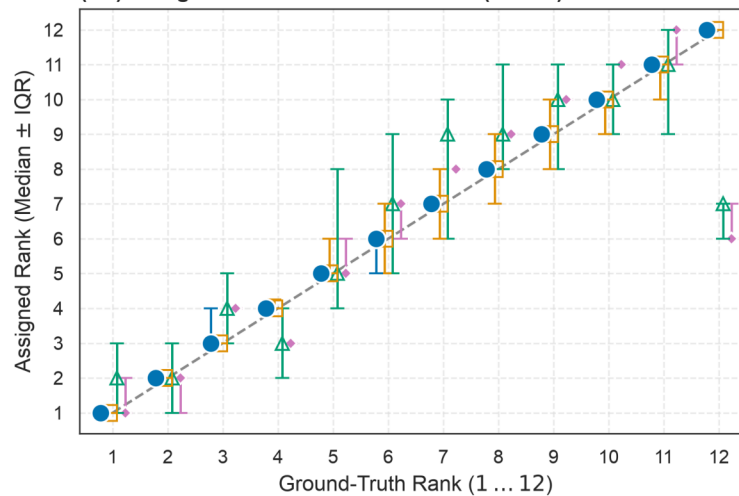
(A3) Rank Dispersion (IQR, $N = 14$)



(B1) Assigned Rank vs. Ground Truth ($N = 10$)



(B2) Assigned Rank vs. Ground Truth ($N = 12$)



(B3) Assigned Rank vs. Ground Truth ($N = 14$)

